

Two Years

What happened to GETTSIM between 2024 and 2026

The Release Log

2023-05-14 v0.7

— 27 months —

2025-08-26 v1.0

2026-01-12 v1.1

2026-03-19 v1.2

2026-08-20 v1.3

2026-09-02 v1.3.1 — yesterday

2024 — Groundwork

2024-01 `soep-preparation` starts as its own repository

2024 Rebuild the infrastructure on `dags.tree` — the tree logic that everything since rests on

2024 Namespaces move to the module level: the taxes and transfers system becomes a nested structure instead of one flat list of names

2024-11 Model classes for policy functions and policy environments; move the project to pixi

No release, because none of it was usable yet on its own.

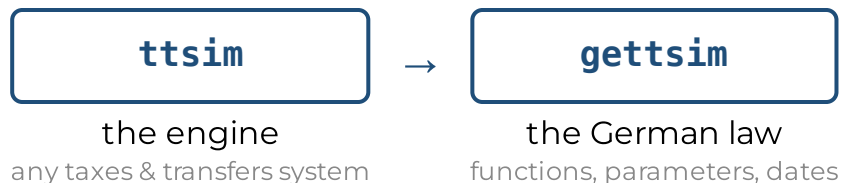
2025 — The Rewrite

One pull request, seven months, merged 2025-07-24.

It was not a refactoring. Almost everything changed:

- the taxes and transfers system became a **tree** of typed objects
- `compute_taxes_and_transfers` became `main` , driven by a DAG of its own
- parameters got real types, a JSON schema, and their own functions
- the whole thing became **vectorized and jittable**

The Split



TTSIM knows about DAGs, aggregation, time conversion, backends, and units. It knows nothing about Germany.

GETTSIM is German law expressed in TTSIM's vocabulary.

The engine has its own test system, METTSIM — a small fictional country. It exists so that TTSIM can be tested without German law getting in the way.

The Offspring



Both exist because of the same complaint: *GETTSIM is great, but I cannot get data into it.*

Personas answer it for a stylised household, **soep-preparation** for our standard data. You will use both tomorrow.

2026 — Correctness (I)

The rewrite bought us the ability to check things.

Static types. `ty` runs over TTSIM and GETTSIM in CI.

Runtime types (GEP 9, June 2026). A package-wide beartype claw. Every `@policy_function` , `@param_function` , `@agg_by_p_id_function` , ... now *requires* an annotation on every argument and on the return value — a missing one is an error at decoration time, not a wrong number three hours later.

Curated errors. A typed exception hierarchy at the user-facing boundaries:

`InputDataError` , `TTTargetsError` , `PolicyFunctionDefinitionError` , ...

When GETTSIM shouts at you this week, that is the point. It is much harder to get a silently wrong number than it was two years ago.

2026 — Correctness (II): Units

GEP 10, accepted July 2026, shipped in v1.3. Units are attached to functions, parameters, inputs, rounding rules, and results — built on Pint.

A `float` column can be wealth, monthly earnings, annual earnings, a share, or square metres. Python is happy to add any of them together. TTSIM is not:

- **€/month vs €/year** — adding a monthly to an annual amount is rejected
- **level** — a household total combined with a Bedarfsgemeinschaft total is rejected
- **currency** — every regime is computed in the currency of its own statute; DM stays DM

The check runs when the policy environment is assembled, by evaluating the formulas with unit-carrying test values.

It does *not* tell you that your formula matches the law. It tells you the arithmetic is not nonsense.

2026 — Substance: SGB XII

Grundsicherung im Alter (Kapitel 4, SGB XII), rewritten June 2026.

- **Einsatzgemeinschaft** replaces the Einstandsgemeinschaft
- **gemischte Bedarfsgemeinschaften** (SGB II next to SGB XII) via the Vertikalmethode
- Freibeträge on pension income from *all three pillars*
- Bedarfsanteilmethode for Bürgergeld / ALG 2
- Grundrentenfreibetrag inside the Wohngeld calculation
- Beitragsbemessungsgrenzen and Durchschnittsentgelte back to 1949

The Interface

What `main` actually is

One Entry Point

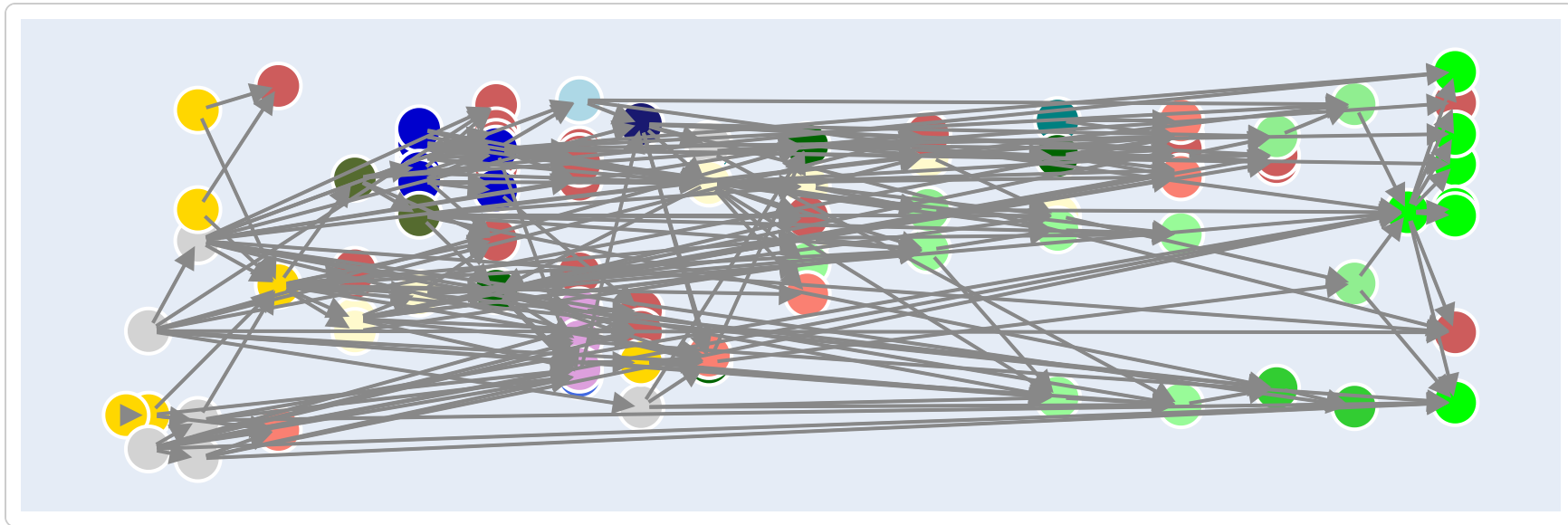
```
from gettsim import InputData, MainTarget, TTTargets, main

outputs_df = main(
    main_target=MainTarget.results.df_with_mapper,
    policy_date_str="2025-01-01",
    input_data=InputData.df_and_mapper(df=inputs_df, mapper=inputs_map),
    tt_targets=TTTargets(tree=targets_tree),
)
```

There is one function. You tell it **which object you want** (`main_target`) and give it the **primitives** that object needs.

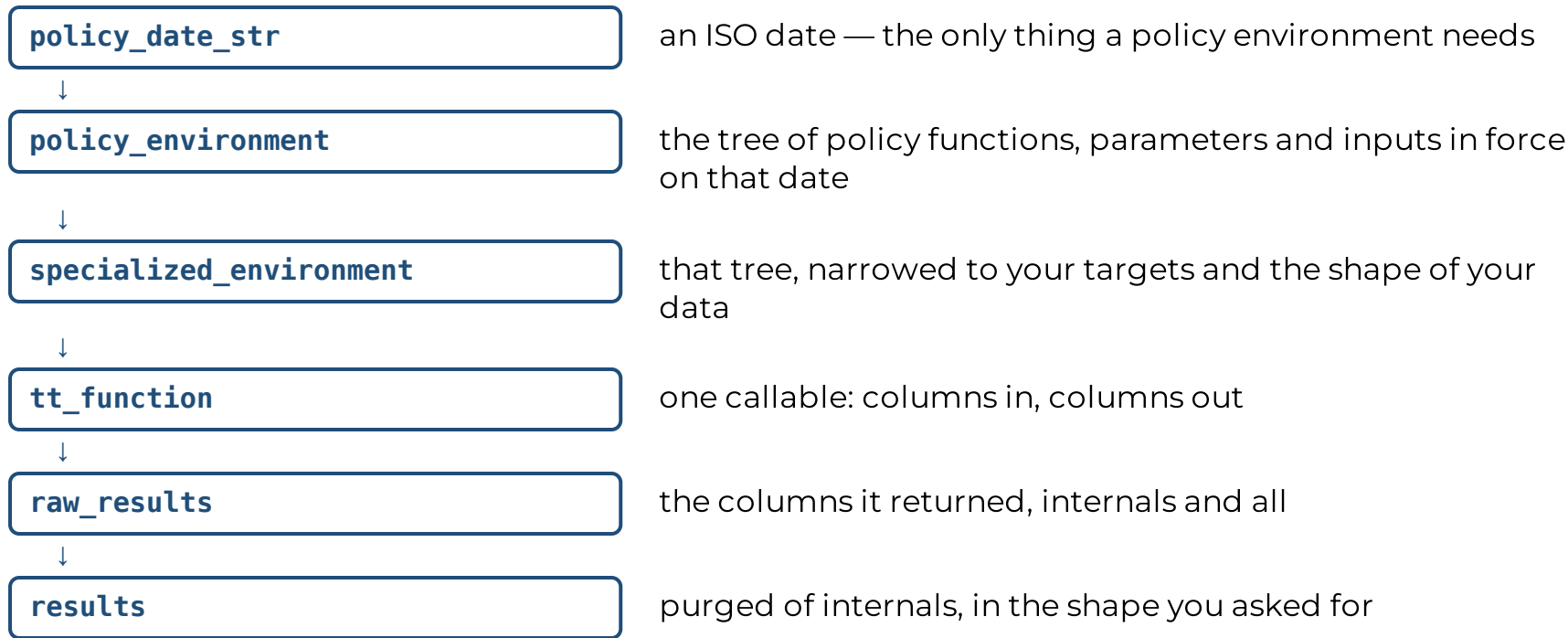
Behind it sits a second DAG — not the taxes and transfers system, but the pipeline that builds and runs it. Ask for any node in it and GETTSIM computes exactly what that node needs, and nothing else.

The Whole Thing



`gettsim.plot.dag.interface()` gives you this in your own session

The Spine



Your data joins at **specialized_environment** , as **processed_data** .

Where Your Data Comes In

`input_data.df_and_mapper`

a DataFrame plus a tree that maps your column names onto GETTSIM's

`input_data.tree / .flat / .qname`

the same data, if you would rather not go through a DataFrame



`input_data_in_computation_currency`

your euros, converted into the statute's currency



`processed_data`

sorted, dtype-canonicalised, ready for the DAG

Every shape ends up in the same place. Pick whichever is least work for you.

Inside `specialized_environment`

`...with_derived_functions`

flatten the tree, then add what can be derived: aggregations to Bedarfsgemeinschaft or Steuernummer level, conversions between month and year — which is why it needs your column names, and therefore your data



`...with_processed_params...`

run the parameter functions; every parameter object becomes a plain number



`...with_partialled_params...`

bake those numbers into the functions — what is left depends only on columns



`tt_dag`

the graph of what remains; compile it and you have `tt_function`

All four are `MainTarget.specialized_environment...` — you can ask `main` for any of them. Useful when a number surprises you and you want to see what the system looked like at that point.

What You Get For Free

`MainTarget.` and let your editor complete. `results`, `templates`, `policy_environment`, `specialized_environment`, `labels`, `unit_checks`, `input_data`, `tt_targets`, `rounding`, `backend`, `fail_if`, `warn_if`, ...

Ask what data you need, instead of guessing:

```
main(  
    main_target=MainTarget.templates.input_data_dtypes.tree,  
    tt_targets=TTTargets.tree(TARGETS),  
    policy_date_str="2025-01-01",  
)
```

Ask for the policy environment, modify it, hand it back. That is Stage 1 this afternoon — and it is why the reform you write today still works tomorrow on personas and on the SOEP.

Two Years

- The engine is separate.
- The law is a tree you can read and change.
- The machine checks your types, your units, and your data.

Now go break it.